

The tough but bright future of SiC MOSFETs

Assignment 6 – EE4C01 Profile Orientation and Academic Skills

Author X & Author Y
October 30, 2019

Abstract— The high-temperature applications ($>200^{\circ}\text{C}$) of metal-oxide-semiconductor field-effect transistors (MOSFETs) are rapidly increasing. The conventional silicon (Si) MOSFETs do not fulfil their demands and Silicon Carbide (SiC) MOSFETs have better thermal properties. The aim of this review is to gain insight into the limited implementation of SiC MOSFETs in high temperature applications, with regard to the technical and financial aspects. In the article comparisons with Si MOSFETs are made, since SiC MOSFETs are currently dominating the MOSFET market. First, a reason why SiC is useful as a high-temperature semiconductor is illustrated. Secondly, the technical aspects of SiC MOSFETs are examined. Research has shown that oxide breakdown, high density of interface states and Fowler-Nordheim-tunneling degrade the reliability of the SiC gate oxide significantly. Thirdly, the financial aspects of the production of SiC MOSFETs are examined. For this, the dominant factors of the total production costs are taken into account. With regard to all the influences on the production costs of high-quality SiC MOSFETs, the quality of the substrate is the most critical factor.

As a conclusion, the limited implementation of SiC MOSFETs in high-temperature applications is due to the reliability of the gate oxide and the costs associated with the production of the substrate.

Index Terms—High-k dielectric materials, high-temperature, production, silicon carbide MOSFET, silicon carbide wafer

I. INTRODUCTION

The electrification of the world is in full swing. The automotive industry has switched from fossil fuels and combustion engines to electricity and electric motors [1],[2]. The increase in electric energy comes at the cost of harsher environments for the electronics inside the described systems. The standard material in fabrication of electronics is silicon (Si) [3]. The demand for higher currents has pushed the working temperature of electronic devices to levels that Si electronics cannot handle.

Regular extrinsic Si, used in integrated circuits, becomes intrinsic at temperatures higher than 200°C due to the thermal generation of electron-hole pairs. Silicon carbide (SiC) outperforms Si in high-temperature applications, because of the wide-bandgap and thermal stability of the material [1]. With the advantages of complementary metal-oxide-semiconductor (CMOS) technology, SiC metal-oxide-semiconductor field-

effect transistors (MOSFETs) are preferred over other SiC devices [2]. The research of SiC MOSFETs began in the late 1980s, and the performances have improved enormously in the last decades [3]. However, the SiC market still covers less than 2% of the total power semiconductor market [4].

The aim of this review is to gain insight into the limited implementation of SiC MOSFETs in high temperature applications, by regarding the technical and financial aspects. Throughout the article some comparisons with Si MOSFETs are made, since SiC MOSFETs are currently dominating the MOSFET market [4].

First, to gain some understanding of the objective, a reason why SiC is useful as a high-temperature semiconductor is formulated. Secondly, the technical aspects of SiC MOSFETs are examined. This is done by addressing the reliability of SiC MOSFETs. Thirdly, the financial aspects of SiC MOSFETs are examined. This is done by going into the dominant factors of the total production costs. Finally, a conclusion is drawn to answer the main question.

II. EXAMPLE OF BETTER PROPERTIES OF SiC: HIGH-TEMPERATURE STABILITY

In MOSFETs, an increase in temperature leads to an increase in their on-state resistance. If a MOSFET conducts significant currents, more heat is generated. At some point, the resistance of the semiconductor material decreases with temperature, allowing even higher currents to flow. This is the point where thermal runaway occurs. From that point on, positive feedback will destroy the MOSFET [3].

At high-temperatures the on-resistance of a SiC MOSFET can be divided into two parts: the channel resistance and the residual resistance. The residual resistance increases when temperature increases, while the channel resistance decreases with increasing temperature. This results in a better high-temperature stability. Measurements show that SiC MOSFETs are capable of reaching a steady-state condition at junction temperatures as high as 250°C . This is approximately 100°C higher than for Si MOSFETs [3].

III. THE TECHNICAL ASPECTS OF SiC MOSFETs

To examine the technical aspects of a MOSFET, this section is divided into three parts. First, the basic structure of a MOSFET is explained. 1ly, the main technical issue of the MOSFET is addressed. Lastly a conclusion of this section is given.

A. Basic MOSFET structure

The base element of a MOSFET is the MOS capacitor. This capacitor consists of two plates; on the one hand the metal gate and on the other the semiconductor material. The two plates are separated with a dielectric, which most of the time is silicon dioxide (SiO_2). Using oxidation processes, the semiconductor material on the bottom reacts with oxygen to form a thin SiO_2 layer. The metal top plate is later attached on top of the oxide.

When a voltage is present at the gate, the electric field creates an inversion layer underneath the oxide. This layer allows current to flow from the drain terminal to the source terminal. See [5] for a more detailed explanation of a MOSFET layout.

B. Main technical issue

Since high electric fields are present across the thin oxide layer, the reliability of the MOS capacitor is critical for long-term use [1], [6]. In the following section, three issues regarding the reliability of the thin oxide layer of a SiC MOSFET are presented. Where possible a solution is given.

1) Oxide breakdown

The first reliability issue in the design of SiC MOSFETs is the breakdown of the oxide layer. In a regular silicon MOSFET, the maximum electric field is limited by the breakdown field of the silicon ($\sim 3 \times 10^5$ V/cm). The SiO_2 layer remains well below its maximum electric field, since the relative permittivity of oxide is about a third of that of silicon. In SiC, however, the maximum electric field is much larger than the breakdown field of SiO_2 . This means the field strength in the oxide layer is reaching the reliability limit [6], [7].

Even more, the threshold voltage for SiC is of an order of magnitude higher than Si, with the same doping concentrations and depletion widths. In order to keep the threshold voltage the same as in regular Si CMOS technology, the oxide thickness has to decrease, which leads to an even worse reliability. Considering these before mentioned problems, the current SiC MOSFET technology has a worse gate oxide reliability than Si MOSFETs [6].

A possible solution to increase the oxide reliability is choosing a different dielectric material. These dielectrics need to have a high relative permittivity in order to accommodate for the higher electric fields. Some examples of high- κ materials are Al_2O_3 and HfO_2 . These materials are currently only in the research stage, but are promising for future applications [8], [9]. A major drawback of high- κ oxide materials is its increase in gate capacitance, since the capacitance is directly proportional to the permittivity of the dielectric.

2) SiC/ SiO_2 interface

A second reliability issue is due to the high amount of imperfections in the SiC/ SiO_2 interface. Regular Si is less chemically inert than SiC, hence it will react faster with O_2 to form the strong oxide layer. This process is well understood

since Si has been used for decades as a semiconductor material. The issue with SiC is that it is more stable than Si. The reaction with O_2 is slow and the presence of carbon in SiC leads to the forming of CO during the oxidation. This gas has to diffuse out of the oxide film [10]. The presence of carbon in the forming of the oxide-semiconductor interface leads to a higher interface state density. When electrons flow from source to drain in the induced channel of the MOSFET, the electrons can be trapped in these so called interface states. This means that the electrons temporarily become bonded to a defect in the lattice, before continuing the journey from source to drain as a free charge carrier. Due to the electron trapping, the high electron mobility of SiC (a key performance metric for semiconductors) is reduced significantly [11].

Again, finding other dielectric materials that chemically bind better to SiC than SiO_2 could be a solution [8], [9].

3) Fowler-Nordheim tunneling

At last, an issue that has to be solved to obtain practical reliability numbers is the effect of Fowler-Nordheim (FN) tunneling. This has to do with the wide bandgap nature of SiC. While the wide bandgap is beneficial for high-temperature electronics, the higher Fermi level of SiC lies closer to the conduction band of the oxide. Compared with Si, the gate leakage current through the dielectric material is higher for SiC at the same temperature and electric field [12]. Over time, the constant injection of charge carriers in the dielectric results in time-dependent dielectric breakdown (TDDB). This is due to the forming of a path of fixed charges, ultimately leading to the breakdown of the dielectric. A high- κ dielectric reduces the electric field across the dielectric. Since FN tunneling is proportionally with the electric field, high- κ materials offer a promising forecast.

C. Concluding the technical aspects

Oxide breakdown, SiC/ SiO_2 interface states and FN tunneling all lead to a worse gate oxide reliability in a SiC MOSFET compared with Si. This prohibits the large scale production of SiC MOSFETs.

It seems that high- κ materials are the perfect solution in future SiC MOSFETs. However, there is no easy way when it comes to choosing a dielectric. Since high dielectric constants usually have a lower electric breakdown field, the use of a dielectric with a high permittivity has to be taken into consideration [13].

IV. THE FINANCIAL ASPECTS OF SiC MOSFETs

To examine the financial aspects of a MOSFET, this section is divided into four parts. First the significance of the large contributors to the total manufacturing costs of SiC MOSFETs are addressed. Secondly, the most significant cost contributor is reviewed. Thirdly, the issue of the main cost contributor is examined. Finally, conclusion of this section is given.

A. The significance of the large contributors

The manufacturing process of a SiC power module is examined in different countries. Fig 1. describes the contribution of all costs associated with the production of SiC MOSFETs per 6-inch wafer [4].

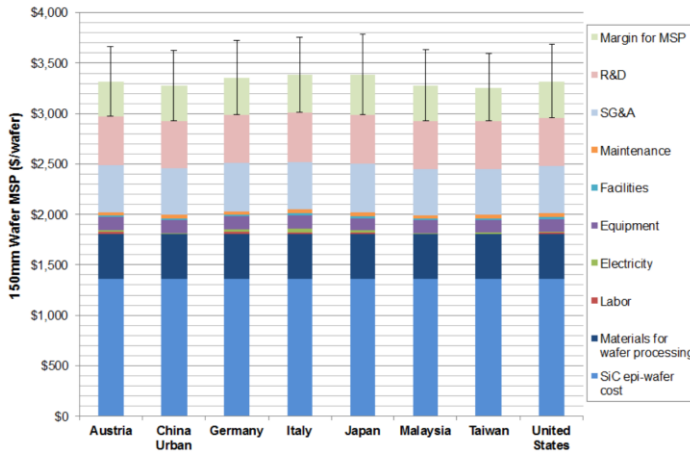


Fig. 1. SiC Wafer MSP [4, Fig. 8]

A breakdown of the most dominant factors in the production costs, based on Fig. 1, is presented. Before going into the most dominant factors, the definitions used in Fig 1. are explained. The MSP (Minimum Sustainable Price) entails a pricing concept for goods and services, in which all manufacturing costs are reflected and with which the investors can be paid back at exactly their minimum required return rate [4]. Note that the MSP is not the market price; it does not take the effects of economic factors into account. The MSP is calculated using a discounted cash flow. The sales, general and administrative (SG&A) as well as the research and development (R&D) costs are calculated as an input in the form of a percentage of the revenue. The SG&A and the R&D costs are high, which may be due to the early stage of the industry. The equipment costs can be decreased by upgrading the volume in which they are made. The large costs of the materials for wafer processing is due to the manufacturing process.

As the profits with MOSFET manufacturing exceed those of the wafer production significantly, and as, at the same time, the MOSFET manufacturing does not represent an important part of the costs, it seems that the SiC epi-wafer costs are decisive and therefore critical [4].

B. Most significant cost contributor

In the last paragraph it becomes clear that the costs of the SiC epi wafer is decisive. In this paragraph, the costs of the manufacturing of a SiC epi-wafer is addressed. To produce a SiC MOSFET, an epi-wafer needs to be made, which then is processed into a MOSFET. An epi-wafer is produced by epitaxially growing SiC on the substrate. To produce a high quality MOSFET a high quality epi-wafer has to be made. To get a high quality of the epi-wafer, the substrate should contain a minimal amount of defects.

In the investigation of the epitaxially growth of SiC on the substrate, it becomes clear that the epitaxial growth causes its own defects in the epi-wafer. However, these defects do not give a major contribution to the total defects of a SiC MOSFET. The majority of the visible defects after the growth are due to the substrate [14]. To compete against the Si MOSFET in the current environment, the high quality of the SiC substrate is of

crucial importance and the production of such a high quality drives up the total costs of the SiC epi-wafer [4].

C. The issue of the main cost contributor

Although many steps in the production of the substrate are the same for SiC MOSFETs as for Si MOSFETs, several steps must be added to the manufacturing procedure of SiC MOSFETs. Examples are the addition of high-temperature anneal, modifications to the metallization and more. These additional steps drive up the price of MOSFETs. Otherwise, the costs for SiC and Si do not differ much [4].

D. Concluding the financial aspects

With regard to all the influences on the production of high-quality SiC MOSFETs, the quality of the substrate is the most critical, because this substrate contributes most significantly to the total MOSFET costs [11].

V. CONCLUSION

The aim of this review was to gain insight into the limited implementation of SiC MOSFETs in high temperature applications, in regard to the technical and financial aspects. The limited implementation of SiC MOSFETs in high-temperature applications is due to the reliability of the gate oxide and the costs associated with the production of the substrate.

The promising characteristics of SiC are reduced by the lower reliability of the gate oxide, which is caused by the breakdown of the thin oxide layer [6], [7], the high density of interface states at the semiconductor-oxide interface [11], Fowler-Nordheim tunneling [12]. The reliability could be improved by properly choosing high- κ dielectric materials [8], [9]. However, while these alternatives are currently being researched, SiC's reliability is still worse than the regular Si MOSFETs.

With regard to all the influences on the production of high-quality SiC MOSFETs, the quality of the substrate is the most critical, because this substrate contributes most significantly to the total MOSFET costs [11]. Therefore, now that the high quality of the substrate is of such huge importance, the SiC MOSFETs are likely to stay expensive in the near future [4].

REFERENCES

- [1] L. C. Yu, G. T. Dunne, K. S. Matocha, K. P. Cheung, J. S. Suehle and K. Sheng, "Reliability Issues of SiC MOSFETs: A Technology for High-Temperature Environments," *IEEE Trans. Device Mater. Rel.*, vol. 10, no. 4, pp. 418-426, Dec. 2010. doi: 10.1109/TDMR.2010.2077295
- [2] Z. Wang et al., "A High Temperature Silicon Carbide MOSFET Power Module with Integrated Silicon-on-Insulator-Based Gate Drive," *IEEE Trans. Power Electron.*, vol. 30, no. 3, pp. 1432-1445, Mar. 2015. doi: 10.1109/TPEL.2014.2321174
- [3] W. Zhou, X. Zhong and K. Sheng, "High Temperature Stability and the Performance Degradation of SiC MOSFETs," *IEEE Trans. Power Electron.*, vol. 29, no. 5, pp. 2329-2337, May 2014. doi: 10.1109/TPEL.2013.2283509
- [4] K. Horowitz, T. Remo, S. Reese, "A Manufacturing Cost and Supply Chain Analysis of Power Electronics Applicable to Medium-Voltage Motor Drives," *National Renewable Energy Lab. (NREL)*, Golden, CO

(United States), United States, 2017-03-24.

- [5] B. Razavi, "Basic MOS Device Physics" in *Design of Analog CMOS Integrated Circuits*, 2nd ed, New York, NY, USA: McGraw-Hill Education, 2017, ch. 2, pp. 744.
- [6] T. Nguyen, A. Ahmed, T. V. Thang and J. Park, "Gate Oxide Reliability Issues of SiC MOSFETs Under Short-Circuit Operation," *IEEE Trans on Power Electron.*, vol. 30, no. 5, pp. 2445-2455, May 2015. doi: 10.1109/TPEL.2014.2353417
- [7] M. Bhatnagar and B. J. Baliga, "Comparison of 6H-SiC, 3C-SiC, and Si for power devices," *IEEE Trans. Electron Devices*, vol. 40, no. 3, pp. 645-655, March 1993. doi: 10.1109/16.199372
- [8] F. Arith, J. Urresti, K. Vassilevskiy, S. Olsen, N. Wright, A. O'Neill, "High Mobility 4H-SiC MOSFET Using a Thin SiO₂/Al₂O₃ Gate Stack" in *Proc. 48th Eur. Solid-State Device Res. Conf.*, Dresden, 2018, pp. 30-33. doi: 10.1109/ESSDERC.2018.8486896
- [9] K. Y. Cheong et al., "Improved Electronic Performance of HfO₂/SiO₂ Stacking Gate Dielectric on 4H SiC," *IEEE Trans. Electron Devices*, vol. 54, no. 12, pp. 3409-3413, Dec. 2007. doi: 10.1109/TED.2007.908545
- [10] P. Deak, J. Knaup, T. Hornos, C. Thill, A. Gali, and T. Frauenheim, "The mechanism of defect creation and passivation at the SiC/SiO₂ interface", *J. Phys. D: Appl. Phys.*, vol. 40, no. 20, pp. 6242--6253, Oct. 2007. doi: 10.1088/0022-3727/40/20/s09
- [11] X. Guo, Q. Xun, Z. Li, S. Du, "Silicon Carbide Converters and MEMS Devices for High-temperature Power Electronics: A Critical Review," *Micromachines*, vol. 10, no. 406, 19 June 2019.
- [12] R. Singh, "Reliability and performance limitations in SiC power devices," *Microelectronics Reliability*, vol. 46, no. 5, pp. 713-730, May 2006. doi: 10.1016/j.microrel.2005.10.013
- [13] J. W. McPherson, J. Kim, A. Shanware, H. Mogul and J. Rodriguez, "Trends in the Ultimate Breakdown Strength of High Dielectric-Constant Materials," *IEEE Trans. Electron Devices*, vol. 50, no. 8, pp. 1771-1778, Aug. 2003. doi: 10.1109/TED.2003.815141
- [14] A. R. Powell, L. B. Rowland, "SiC Materials-Progress, Status, and potential roadblocks," *Proceedings of the IEEE*, vol. 90, no. 6, pp. 942-955, June 2002.